

Maxwell Lokshin

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Computer Science

Jul. 2023 – May 2027

- Relevant Coursework: Algorithms, Machine Learning, Artificial Intelligence, Computer Vision, Systems Software, Linear Algebra, Discrete Mathematics
- Activities: IEEE @ UCF – SECBOT Software Member; AIRACERS @ UCF – Project Lead; AI @ UCF – Member

Palm Beach State College

Lake Worth, FL

Dual Enrollment, Mathematics

Aug. 2022 – May 2023

EXPERIENCE

Software Engineer

Jul. 2025 – Present

PeakActivity

Boynton Beach, FL (Remote)

- Consult directly with clients to understand technical requirements and contribute to backend solution design
- Build and maintain backend systems applying industry best practices under the guidance of senior engineers
- Learn and apply new coding practices across projects to strengthen full-stack development skills

Project Lead

Mar. 2026 – Present

UCF AIRACERS (F1TENTH)

Orlando, FL

- Led software, electrical, and mechanical teams to deliver a custom autonomous race car in 3 weeks; car completed 9 laps in 5 minutes with a fastest lap of 24 seconds
- Developed a hybrid pure pursuit and dual wall-following algorithm with dynamic mode switching; implemented auto-braking and dual wall detection for self-correction
- Configured Gym-ROS with RViz in Docker for pre-deployment simulation and integrated VESC motor messages on the Nvidia Jetson TX2 for real-time vehicle control

Software Member

Oct. 2025 – Present

IEEE UCF – SECBOT

Orlando, FL (Hybrid)

- Implemented a D* Lite pathfinding algorithm via NAV2 and a perception-to-goal conversion system for adaptive autonomous navigation across a mission field
- Developed an OpenCV computer vision pipeline for real-time object detection, integrated with the autonomous navigation and signal relay workflow
- Integrated a custom UWB ROS2 plugin into Gazebo for field-relative localization; worked across the full ROS2 Jazzy simulation-to-field pipeline

PROJECTS

SECBOT | C++, Python, ROS2, OpenCV, Gazebo, Docker, Git

Sep. 2025 – Present

- Designed and validated D* Lite path planning in C++ using a grid-based map for efficient field traversal
- Built real-time HSV color-tuning GUI enabling interactive calibration of detection parameters without code changes

Nice Green Machine – F1TENTH | Python, ROS2, RViz, Docker, Git

Mar. 2026 – Present

- Built hybrid pure pursuit / dual wall-follow algorithm with dynamic mode switching; achieved 9 laps in 5 min at UCF F1TENTH competition
- Set up Gym-ROS simulation in Docker with RViz for pre-hardware validation and integrated VESC messages on Nvidia Jetson TX2

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, SQL (Postgres), HTML/CSS

Frameworks & Middleware: ROS2, NAV2, React, Flask, FastAPI

Tools & Platforms: Docker, Git, Gazebo, RViz, Nvidia Jetson TX2, VS Code, PyCharm

Libraries: OpenCV, pandas, NumPy, Matplotlib, Pygame